

Suriname Mangrove Climate Assessment

Daily climatology of rainfall, radiation, cloud cover, and convective precipitation, using ERA5

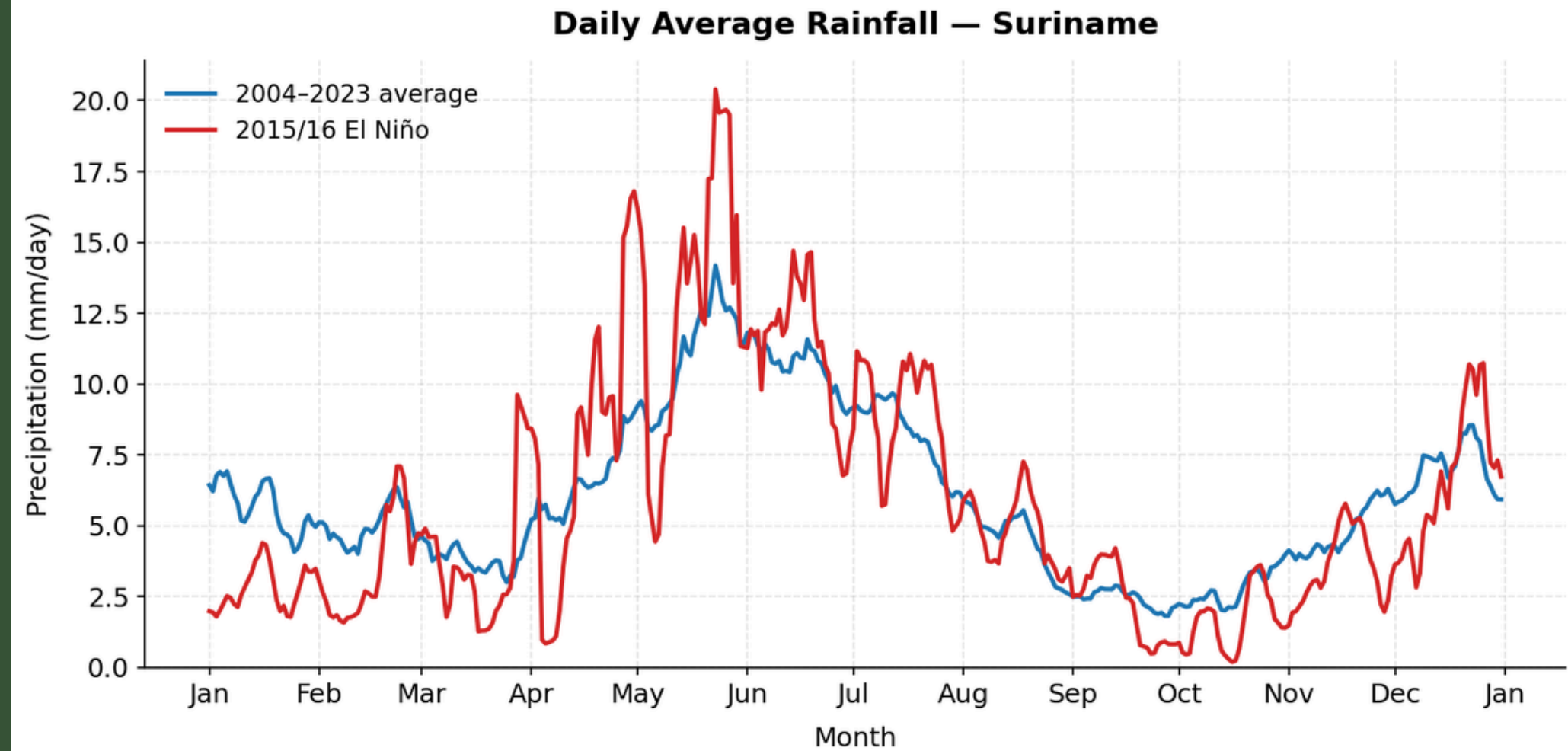
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Data & Methods

- **Data source:** ERA5 reanalysis (fifth-generation atmospheric reanalysis, ECMWF/Copernicus Climate Change Service), accessed via Google Earth Engine
- **Region:** Suriname mangrove coastline, bounding box approximately 5.5–6.25°N, 58–54°W
- **Climatology period:** Daily values averaged across 2004–2023 (20 years), shown in blue
- **Comparison period:** 2015/2016 El Niño year, shown in red
- **Convective precipitation likelihood metric:** Defined as the ratio of convective precipitation to total precipitation each day (convective ÷ total). This indicates what fraction of daily rainfall is driven by localized convective storms versus larger-scale frontal/stratiform systems — a higher value indicates rainfall is more likely to arrive as intense, short-duration convective events rather than steady, prolonged rain.
- All daily series are smoothed with a 7-day rolling average to reduce day-to-day noise while preserving seasonal patterns.

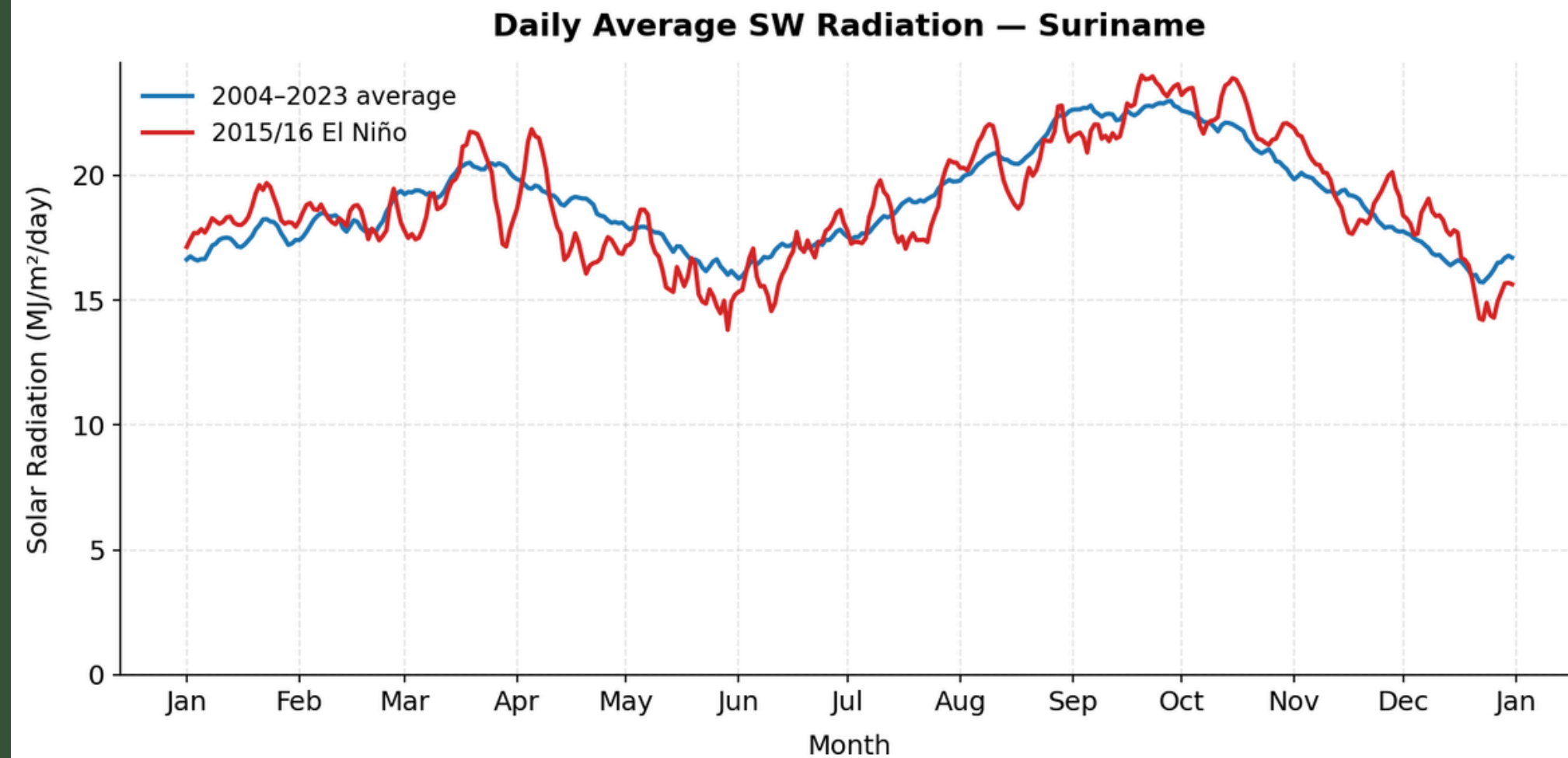
Rainfall

The 20-year average shows Suriname's characteristic bimodal rainfall pattern: a short dry season around February–March, a wetter period from April through July, and a longer dry season from August through November. During the 2015/16 El Niño year, rainfall was markedly more erratic, with several sharp spikes (some days exceeding 50 mm/day) during what is normally the wetter season, alongside periods that dipped below the long-term average — consistent with El Niño's known disruption of typical tropical rainfall timing.



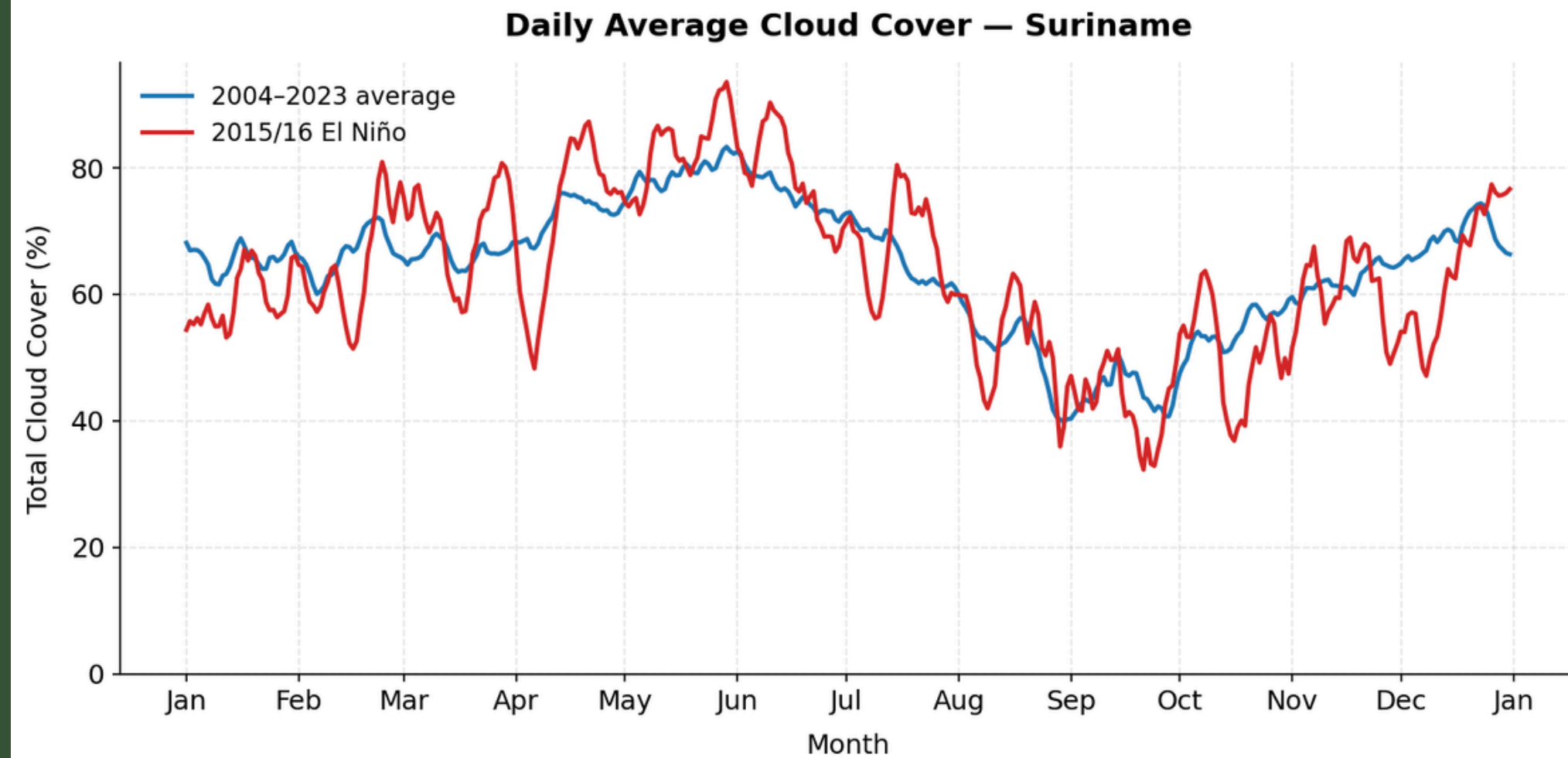
SW Radiation

Solar radiation follows an inverse relationship with rainfall and cloud cover, as expected — radiation is lowest during the wetter months (April–July) and highest during the drier September–October period. The El Niño year shows more day-to-day volatility than the long-term average, with radiation swinging both higher and lower than typical on a given day, again reflecting a disrupted, less predictable seasonal cycle.



Cloud Cover

Cloud cover tracks closely with the rainfall pattern, peaking during the wetter months and dropping during the dry seasons. The 2015/16 El Niño year shows greater extremes in both directions — notably lower cloud cover than average during parts of the normally cloudier wet season, and higher spikes during the typically clearer dry season.



Convective Precipitation Likelihood

The convective fraction remains high year-round (roughly 0.6–0.9), consistent with the tropics, where most rainfall originates from convective storm activity rather than large-scale frontal systems. The fraction rises further during the drier months (approaching 0.9), indicating that the rainfall which does occur during dry periods is even more likely to be short, intense convective activity rather than steady rain. The El Niño year closely tracks the long-term average for this metric, suggesting El Niño's main effect was on rainfall amount and timing rather than fundamentally changing the type/mechanism of precipitation.

